

Multi-Interface nRF52833 Embedded Controller Board

The All Protocol PCB is a compact embedded monitoring and data-acquisition platform based on the Nordic nRF52833 ARM Cortex-M4F microcontroller. It is designed to interface with industrial, agricultural and environmental sensors for real-time data acquisition, processing and local monitoring. The board supports RS485 with Modbus RTU, RS232, UART, I²C, SPI, GPIO and USB-C connectivity, along with battery operation, charging, voltage regulation, protection circuitry and display interfacing in a single PCB.

Key Features

- Nordic nRF52833 ARM Cortex-M4F microcontroller, 64 MHz
- RS485 industrial communication interface with Modbus RTU support
- RS232 serial communication interface for legacy and industrial equipment
- UART, I²C, SPI and GPIO interfaces for sensors, displays and peripheral expansion
- USB-C connectivity for development, programming, testing and board power
- Real-time sensor data acquisition, validation, processing and local visualization
- 2.42-inch OLED display interface for direct monitoring of measurements
- Rechargeable 18650 Li-ion battery operation with external charging support
- Buck-boost voltage regulation for stable system operation across battery conditions
- Power switch, reset control and power-status indication
- Transient-voltage and communication-line protection for improved robustness
- Compact, modular PCB architecture with expansion provisions for additional sensors and peripherals

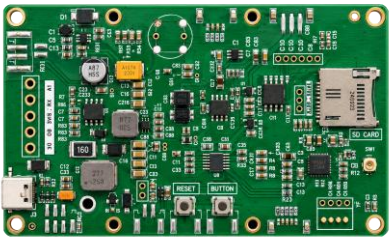


Fig: All Protocol PCB

Technical Specifications

Parameter	Value
Microcontroller	Nordic nRF52833-QDAA
Processor	ARM Cortex-M4F
Clock Frequency	64 MHz
Flash Memory	512 KB
RAM	128 KB
MCU Operating Voltage	1.7 V – 5.5 V
Industrial Interface 1	RS485
Industrial Interface 2	RS232

Industrial Protocol	Modbus RTU over RS485
RS485 Transceiver	MAX13487
RS232 Interface	Dedicated RS232 serial communication interface
Controller Interfaces	UART, I ² C, SPI, GPIO
USB Connectivity	USB-C
Display	2.42-inch OLED
Display Controller	SSD1309
Display Resolution	128 × 64 pixels
Display Interface	I ² C
Battery	Rechargeable 18650 Li-ion
Battery Rating	3.7 V, 2600 mAh
Voltage Regulation	TPS63070 synchronous buck-boost + 3.3 V regulation
Charging	External VIN charging; DC jack/solar charging support in design
Protection	Reverse-polarity, transient/TVS and communication-line protection
User Controls	Power switch and reset button
Status Indication	Power/status LED
Hardware Provisions	SD card slot, U.FL antenna connector and expansion headers

Applications

- RS485 / Modbus RTU sensor interfacing and monitoring
- RS232 industrial and legacy device interfacing
- Agricultural and soil sensor monitoring
- Environmental monitoring and field testing
- Industrial sensor and instrument interfacing
- Laboratory measurement and sensor evaluation
- Real-time embedded data acquisition systems
- Research, education and embedded-system development
- Portable field-monitoring equipment
- Multi-protocol sensor gateway and prototype development